

Average rate of change

Average rate compares an output change with the corresponding input change. Use $a < b$.

$$\frac{f(b) - f(a)}{b - a}$$

What are the units if input is time and output is distance?

A rate from data

A tank contains 80 liters at $t = 2$ minutes and 35 liters at $t = 7$ minutes.

What is the average rate of change over this interval?

A rate from a formula

$$f(x) = x^2 + 1$$

What is the average rate of change from $x = 1$ to $x = 4$?

Compare two intervals

$$f(x) = x^2$$

Is the average rate from 0 to 2 the same as from 2 to 4?

A constant rate

$$f(x) = -3x + 8$$

What is its average rate of change over any interval?

The secant line

A secant line passes through two points of a graph.

How is its slope related to average rate of change?

A zero average rate

$$f(x) = x^2$$

Does the average rate from -2 to 2 show that the function is constant?

A symbolic rate

Use $h \neq 0$ and $f(x) = x^2 + 3x$.

$$\frac{f(x + h) - f(x)}{h}$$

What simplified expression gives this rate?

Increasing on an interval

A function increases on an interval when larger inputs give larger outputs.

Is an increasing function required to have positive outputs?

Decreasing on an interval

A function decreases on an interval when larger inputs give smaller outputs.

Could a decreasing function have positive outputs throughout an interval?

Constant on an interval

A function is constant on an interval when all inputs there give the same output.

What does this part of its graph look like?

Intervals from a parabola

$$f(x) = (x - 2)^2 - 4$$

On which intervals is f decreasing or increasing?

A graph with straight segments

Connect $(-3, 0)$, $(-1, 4)$, $(2, 4)$, and $(5, -2)$ in order with straight segments.

On which intervals is this function increasing, constant, or decreasing?

A local maximum

At a local maximum, the function value is at least as large as nearby function values.

Does a local maximum have to be the largest value on the entire domain?

A local minimum

At a local minimum, the function value is no larger than nearby function values.

What change in direction often appears at a local minimum?

Locate local extrema

Use a graphing utility.

$$f(x) = x^3 - 3x$$

Where are the local maximum and local minimum?

An absolute maximum

An absolute maximum is a function value at least as large as every value on its domain.

Can a function have a local maximum without an absolute maximum?

An absolute minimum

An absolute minimum is a function value no larger than every value on its domain.

$$f(x) = (x + 1)^2 + 3$$

What are the minimum value and the input where it occurs?

An extremum on a restricted domain

Use $-1 \leq x \leq 3$.

$$f(x) = x^2$$

What are the absolute minimum and maximum values?

An open endpoint

Use $0 < x < 2$.

$$f(x) = x$$

Does this function attain an absolute minimum or maximum?

Local versus absolute

Use a graphing utility on the domain $-3 \leq x \leq 3$.

$$f(x) = x^3 - 3x$$

Is its local maximum also its absolute maximum?

A viewing-window limitation

A calculator graph appears to have a highest point near the edge of the window.

What evidence is needed before calling it an absolute maximum?